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THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Kaprekar Contest

(NMTC- at **SUB-JUNIOR LEVEL** *VII to VIII Standards)*

Saturday, 25^{th} August 2007.

Note:

1) *Fill in the answer sheet with your Name, Class, and Section etc. in the specified places.*
2) *Only one of the choices A,B,C,D is correct for each question. Write the alphabet of your choice in the answer sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).*
3) *For each correct response you get 1 mark;* ***for each incorrect response you lose ¼ mark.***
4) *Diagrams are only visual aids; they are not drawn to scale.*
5) *You are free to do rough work on separate sheets.*
6) *Duration of the test: 2 p.m. to 4. p.m – 2hours.*

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1. a, b, c, d are reals such that $a - 2005 = b + 2006 = c - 2007 = d + 2008$. The greatest among a, b, c, d is

   (A) a **(B)** b **(C)** c **(D)** d

2. The number of prime less than 100 which have 7 as the unit digit is

   (A) 6 **(B) 7** **(C) 8** **(D) 9**

3. A student got x marks in a test. The student who got the first mark gets 48 more than this student who got x marks. If the total marks of both the students is 110. the highest mark secured is

   (A) 83 **(B) 92** **(C) 79** **(D) 100**

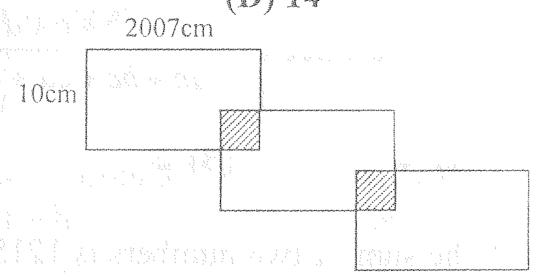
4. In a triangle if one angle is greater than the sum of the other two angles then the triangle is

   (A) Acute angled (B) Right angled (C) Obtuse angle (D) Equilateral

5. The largest positive integer 'n' for which $n^{200} < 6^{300}$ is

   (A) 12 **(B) 13** **(C) 17** **(D) 14**

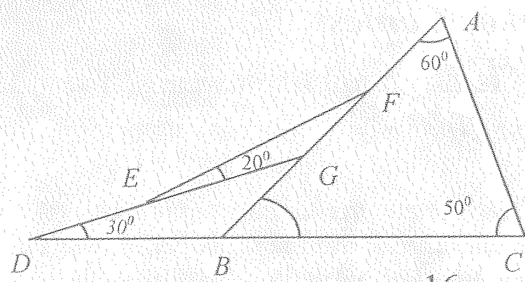
6. Three identical rectangles are overlapping as in the diagram. The length and breadth of each rectangle are respectively 2007cm and 10cm. The area of each of the shaded square portions is $16cm^2$. The perimeter of the outer boundary of the figure in cm is



   (A) 10070 **(B) 12070** **(C) 14070** **(D) 11070**

7. 20% of 50% is what percent of 25% of 40%

(A) 80% (B) 60% (C) 65% (D) 100%

8. 

In the adjoining figure $A = 60^0$, $C = 50^0$. $\angle BDG = 30^0$ $\angle GEF = 20^0$. Then

(A) $EG = 2FG$ (B) $EG > FG$
(C) $EG = FG$ (D) $EG < FG$

9. A boy on being asked what $\frac{16}{17}$ of a fraction was made the mistake of dividing the fraction by $\frac{16}{17}$, and got an answer which exceeded the correct answer by $\frac{33}{340}$. The correct answer is

(A) $\frac{64}{85}$ (B) $\frac{65}{84}$ (C) $\frac{32}{17}$ (D) None of these

10. x,y,z are three sums of money such that y is the simple interest on x and z is the simple interest on y for the same time and rate then

(A) $x^2 = yz$ (B) $y^2 = zx$ (C) $z^2 = xy$ (D) None of these

11. How often the hands (hour hand and minute hand) of a clock are in a straight line every day?

(A) 2 (B) 4 (C) 16 (D) 44

12. It is given that $\frac{x}{y} = \frac{4}{5}$, which one of the following is incorrect

(A) $\frac{x+y}{y} = \frac{9}{5}$ (B) $\frac{y+2x}{x} = \frac{13}{4}$ (C) $\frac{x^2+y^2}{xy} = \frac{41}{20}$ (D) $\frac{2x^2-y^2}{xy} = \frac{9}{20}$

13. If a,b,c,d are positive integers such that $a = bcd$, $b = cda$, $c = dab$ and $d = abc$, then the value of $\frac{(a+b+c+d)^4}{(ab+bc+cd+da)^2}$ is

(A) 4 (B) 2 (C) 1 (D) $\frac{1}{2}$

14. The sum of two numbers is 1215 and their G,C,D is 81. How many pairs of such numbers are possible?

(A) 2 (B) 4 (C) 6 (D) 8

15. If $60^a = 3$ and $60^b = 5$ then the value of $12^{\frac{1-a-b}{2(1-b)}} = ?$

(A) $\sqrt{60}$ (B) $\sqrt{3}$ (C) 5 (D) 2

16. The number of digits in $8^{15}\, 5^{40}$ (where written in base 10 form) is

(A) 42 (B) 45 (C) 55 (D) 2007

17. In a school, there are 5 times as many boys as girls, and 6 times as many girls as teachers. If b,g,t represent the boys, girls and teachers respectively the total number of boys, girls and teachers in the school is

(A) 37 b (B) $\frac{37}{30}b$ (C) 30 g (D) 37 g

18. Consider the equation $x^2 + y^2 = 2007$. Given x is a real number and y is a natural number, the number of solutions of the equation is

(A) 0 (B) 2006 (C) 88 (D) 44

19. The number of two digit numbers whose digit sum is divisible by 6 is

(A) 13 (B) 8 (C) 7 (D) 22

20. A number when divided by 899 gives a remainder 63. What is the reminder when the number is divided by 29.

(A) 5 (B) 28 (C) 16 (D) 12

21. A man gets $\left(\frac{a}{b}\right)^{th}$ of Rs.100 and $\left(\frac{b}{a}\right)^{th}$ of Rs.100 again. He gives away Rs.200.

Then the man

(A) loses by the transaction

(B) will not lose by the transaction

(C) loss or gain depends where $a > b$ or $a < b$ respectively

(D) loss or gain depends where $a < b$ or $a > b$ respectively

22. A six digit number is formed by repeating a three digit number twice (like 245245). Such numbers are always divisible by

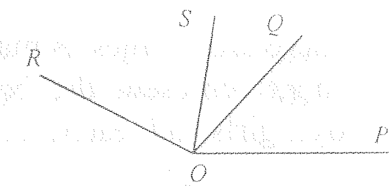
(A) 1001 (B) 25 (C) 101 (D) 111

23. Twenty four children are seated equally spaced around a circle and numbered from 1 to 24. what is the number of the child who sits diametrically opposite to the child number 10

(A) 21 (B) 23 (C) 22 (D) 20

24. The lengths of the altitudes of a triangle are in the ratio 1:2:3. Then
(A) one angle of the triangle must be 60^0
(B) the triangle is a right angled triangle
(C) the triangle is an obtuse angled triangle
(D) such a triangle does not exist

25. OP, OQ, OR are rays OS bisects $\angle ROP$, $\angle POQ = 62^0$, $\angle ROQ = 102^0$. Then $\angle SOQ =$



(A) 20^0 **(B) 15^0** **(C) 25^0** **(D) 10^0**

26. If x, y, z, a, b, c are real and none of these quantities is zero $xy = a$, $xz = b$ and $yz = c$ then $x^2 + y^2 + z^2 =$

(A) $a^2 + b^2 + c^2$ (B) $\dfrac{a^2 + b^2 + c^2}{abc}$

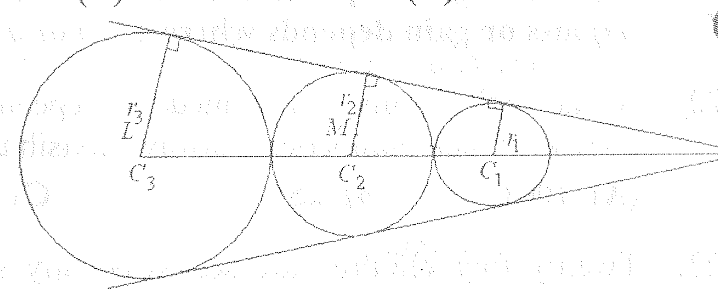
(C) $\dfrac{abc}{a^2 + b^2 + c^2}$ (D) $\dfrac{(ab)^2 + (bc)^2 + (ca)^2}{abc}$

27. The perimeter of a right angled triangle is 144 cm and the hypotenuse is 65 cm. Its area in square cm is
(A) 506 **(B) 508** **(C) 1440** **(D) 504**

28. The number $A7389B$ where A,B are digits is divisible by 72. then (A, B) is
(A) (6,3) **(B) (3,6)** **(C) (4,7)** **(D) (7,5)**

29. On the sale of 10m cloth, a gain equal to the selling price of 2m cloth was obtained. The gain is
(A) 20% **(B) 10%** **(C) 25%** **(D) 40%**

30. Three Circles C_1, C_2, C_3 with radii r_1, r_2, r_3 where $r_1 < r_2 < r_3$ are placed as in the figure. Then $r_2 =$



(A) $\sqrt{r_3 - r_1}$ (B) $\sqrt{r_3 + r_1}$ (C) $\sqrt{r_3^2 + r_1^2}$ (D) $\sqrt{r_3 \cdot r_1}$